Mobile Product Analytics Portfolio • Automated Daily Briefing • *2026-08-22*

■ **Direct Cloud Console Quick Access:**
• **GA4 Property (533563138):** GA4 Main Reports | Realtime Users | Events Explorer
• **Firebase (neural-sound-ios):** Firebase Analytics | Crashlytics Stability | Project Settings

Portfolio Analytics & Executive Growth Architecture Report

Prepared by: Principal Mobile Analytics & Growth Architect
Data Scope: Cross-App Firebase / GA4 Daily & Historical Multi-Horizon Intelligence
Reporting Date: August 22, 2026 (Dataset Snapshot: August 15 – August 21, 2026)

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1. Portfolio Executive Summary & Health Dashboard

Cross-App Portfolio Benchmark Table

Metric / Horizon	Music Separation iOS (com.neuralsound.app)	Vocal Cleaner (com.neuralsound.audioclean)	Portfolio Total / Composite
Yesterday Active Users (DAU)	143	3	**146**
Yesterday New Users	107 (74.8%)	0 (0.0%)	**107 (73.3%)**
Yesterday Sessions	169	17	**186**
Engagement Rate (Yesterday)	84.6%	88.2%	**84.9%**
Avg Session Duration	5m 33s (333.1s)	58m 51s (3,531.1s)*	**10m 32s (Composite)**
Crash-Free User Rate	**100.0%** ■	**66.67%** ■	**99.32%**
Past 7 Days Users (WAU)	419	11	**430**
Past 30 Days Users (MAU)	841	27	**868**
Lifetime Active Users	1,056	27	**1,083**
Top Growth Market	South Korea (76.9%)	Bangladesh (100.0%)	South Korea / Bangladesh
Primary Risk / Bottleneck	Slider Event Flooding & Low D7 Retention	Critical App Crash (33.3% rate) & StoreKit Loop	Technical Stability & Telemetry Noise

**Note: Vocal Cleaner's extreme session duration is skewed by background recording/audio processing timers remaining active without user termination.*

Portfolio Health & Cross-App Trends Synthesis

1. **Asymmetric Portfolio Scale & Hyper-Growth Spike**: *Music Separation iOS* accounts for **97.9% of portfolio DAU** and **97.4% of WAU**. The app experienced a massive acquisition velocity spike starting August 18, growing DAU by **376% in 4 days** (from 30 to 143 daily users). Over **30% of its lifetime user base** was acquired in the last 7 days alone, heavily concentrated in **South Korea**.
2. **Critical Telemetry Infrastructure Defects**: Both applications suffer from implementation flaws in logging:
3. **Music Separation iOS**: Un-debounced UI sliders (`player_pitch_changed`, `player_tempo_changed`) generate up to **570 events per user**, threatening to hit GA4 event volume limits and artificially inflating events-per-session (97.8 events/session).

4. **Vocal Cleaner**: A single user generated **12** `purchase_started`, **10** `purchase_approved`, **6** `purchase`, and **5** `purchase_failed` events in 24 hours, pointing to a payment retry loop or broken transaction state machine.
5. **Severe Quality & Stability Divergence**: While *Music Separation iOS* maintains a **100% crash-free user rate**, *Vocal Cleaner* exhibits a dangerous **66.67% crash-free rate** (1 out of 3 users crashed yesterday), rendering the build unviable for user acquisition until patched.

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2. Deep-Dive: Music Separation iOS

App Header & Health Score

- **Bundle ID**: `com.neuralsound.mss` | **GA4 Property ID**: `533563138` | **Stream ID**: `14402502632`
- **Health Score**: **8.4 / 10**
- **Rationale**: Strong acquisition momentum (+376% 7d growth), high daily engagement rate (84.6%), and 100% crash-free stability. Deductions applied for: (1) Severe UI slider telemetry event flooding bug, (2) Low absolute returning user volume (only 41 returning DAU vs. 107 new), (3) Extreme market single-point concentration risk (South Korea = 76.9% DAU), and (4) Engagement regression in the new staged rollout version (`1.3.08`).

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon Window	Active Users (AU)	New Users (NU)	Total Sessions	Engagement Rate	Growth / Trajectory Dynamics
Yesterday (Latest 24h)	143	107	169	84.6%	**Peak Velocity**: 74.8% of daily users are new installs.
Past 7 Days (WAU)	419	311	595	84.5%	**Hyper-Growth Spike**: 29.4% of total lifetime users joined this week.
Past 30 Days (MAU)	841	708	1,648	85.6%	**Explosive Expansion**: MAU comprises 79.6% of all-time installs.
Past Year (365 Days)	1,056	933	2,487	84.8%	**Recent Inception**: Entire app lifecycle is contained within recent months.
All-Time / Lifetime Total	1,056	933	2,487	84.8%	**Baseline Scale**: 1.05k Total Lifetime Installs.

Velocity, Acquisition Trajectory & Retention Dynamics Analysis

- **Acquisition Velocity**: The app is undergoing early-stage viral discovery. The 7-day run-rate (419 WAU) represents **49.8% of the 30-day active base**, proving that acquisition velocity has accelerated dramatically within the last 72 to 96 hours.
- **Retention Deficit**: While acquisition is spiking, lifetime retention depth remains shallow. Out of 1,056 all-time users, 30-day active users stand at 841, but 7-day active users drop to 419. Only **41 returning users** accessed the app yesterday, showing that despite high initial session engagement (84.6%), early user retention (D1-D7) requires optimization.

Old vs. New User Cohort Deep-Dive

User Cohort Type	Active Users	Sessions	Avg Session Duration	Engagement Rate	Screen Views	Key Behavioral Insights
New Users	114	112	347.8s (5m 47s)	**94.6%**	229	High initial exploration; complete onboarding and run core separation audio processing.
Returning Users	41	52	332.9s (5m 32s)	**69.2%**	39	Lower engagement rate (-25.4% vs new). High utility focus; users perform targeted edits and exit.
Unattributed / `(not set)`	0	5	6.5s	20.0%	1	Ghost background executions or background fetch events.

- **Cohort Insight**: New users display extraordinary initial engagement (**94.6%**), validating product-market fit during first-touch experiences. Returning users show a **25.4 percentage-point drop in engagement rate (69.2%)**, indicating that utility drops after initial file separation or users experience workflow fatigue.

7-Day Velocity & Trajectory

Date (YYYY-MM-DD)	Active Users (DAU)	New Users	Total Sessions	Engagement Rate	Daily Trend Notes
2026-08-15	30	4	46	82.6%	Baseline steady state. High sessions/user ratio (1.53).
2026-08-16	28	7	38	84.2%	Organic weekend stability.
2026-08-17	36	14	45	80.0%	Early acquisition uptick (+100% new users DoD).
2026-08-18	57	36	71	83.1%	Expansion inflection point (+58.3% DAU DoD).
2026-08-19	94	78	120	**90.8%**	Korean viral acquisition surge begins. Peak engagement.

2026-08-20	93	65	113	82.3%	Mid-week plateau / user absorption phase.
2026-08-21	**143**	**107**	**169**	84.6%	**All-Time Peak DAU**: Driven by organic growth in South Korea.

User Funnel Diagrams

Onboarding Conversion Funnel

```
[109] first_open (100.0%) ■ ▼ 100.0% Conversion [111] onboarding_viewed / onboarding_step_viewed ■ ▼ 99.1% Conversion [110] onboarding_next_tapped ■ ▼ 100.0% Conversion (117 viewed UI) [110] login_succeeded ■ ▼ 96.4% Conversion [106] first_journey_login_completed (Overall Onboarding Funnel Efficiency: 97.2%)
```

- **Onboarding Funnel Analysis**: Exceptional funnel performance. **97.2% of users** who open the app complete onboarding and successfully authenticate. Friction at onboarding is virtually zero.

Core Feature Execution Funnel (Import to Audio Separation)

```
[106] Onboarded & Authenticated Users ■ ▼ 87.7% Click Action [93] first_journey_add_separation_clicked / plus_button_tap [105] ■ ▼ 50.5% Import Start Rate <-- ■ MAJOR DROP-OFF POINT (-49.5%) [53] import_started ■ ■■■ [78 Users] Files App Import ■ ■■■ [63 Users] Photos Import ■ ▼ 88.7% Processing Success Rate [47] separation_play_started (Overall Core Conversion from Intention: 44.8%) ■ ▼ 153.2% Re-engagement / Player Loop [72] separation_player_opened -> [20] Pitch/Tempo Player Opened
```

- **Core Loop Bottleneck**: The primary product drop-off occurs between clicking "Add Separation" / `plus_button_tap` (105 users) and actually executing `import_started` (53 users). **49.5% of users abandon the workflow when the OS File/Photo Picker native sheet opens**, likely due to permission prompts, unsupported file formats, or lack of sample audio files.

Crash & Stability Health

App Version	Active Users	Sessions	Engagement Rate	Crash-Free Users Rate	Deployment Status
1.3.07	136	152	**86.8%**	**100.0%**	Production Baseline (95.1% Traffic)
1.3.08	5	13	**53.8% ■■**	**100.0%**	Active Staged Rollout (3.5% Traffic)
1.3.06	2	4	100.0%	100.0%	Legacy Version (1.4% Traffic)

- **Stability Status**: Excellent absolute stability with **100.0% Crash-Free Users** across all active builds.

■ New Build / Staged Rollout Alert Section

```
[STAGED ROLLOUT WARNING: BUILD 1.3.08]
-------------------------------------------------------------------------------- Baseline (v1.3.07) Engagement Rate : 86.8% (136 Users, 152 Sessions) New Build (v1.3.08) Engagement Rate: 53.8% (5 Users, 13 Sessions) Delta Shift : -33.0 Percentage Points (Significant Drop) Crash-Free Users Rate : 100.0% (No raw crashes reported)
```

```
--------------------------------------------------------------------------------
```

- **Architectural Risk Assessment**: Build `1.3.08` shows a **33.0 percentage-point collapse in Engagement Rate** compared to `1.3.07`. While the sample size is small (5 users, 13 sessions), the lower engagement rate indicates potential non-fatal bugs, UI regressions, broken playback state machines, or audio pipeline hangs that stall user activity without triggering a crash. **Halt staged rollout expansion until build 1.3.08 UX telemetry is verified.**

Top In-App Events & Feature Lookups

Critical Telemetry Anomaly: Un-Debounced UI Sliders

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TELEMETRY ANOMALY DETECTED: EVENT FLOODING BUG ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ Event
Name ■ Total Events ■ Unique Users ■ Events / User ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
player_pitch_changed ■ 5,706 ■ 10 ■ 570.6 ■ EXTREME ■ ■ player_tempo_changed ■ 2,046 ■ 9 ■ 227.3 ■ HIGH ■ ■
player_stem_balance_changed ■ 132 ■ 1 ■ 132.0 ■ HIGH ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

- **Technical Root Cause**: The pitch, tempo, and stem balance UI sliders trigger analytics logging directly on the UI slider `valueChanged` continuous event (e.g., `UISlider.valueChanged` or SwiftUI `Binding` update) rather than logging on touch release / editing end (`touchUpInside` / `onEditingChanged`).
- **Impact**: 10 users generated **7,884 slider events** (out of ~16,500 total daily events for the entire app). This pollutes GA4, inflates session event counts to 97.8 per session, risks breaching GA4 daily event logging limits, and artificially distorts engagement telemetry.

Core Functional Event Distribution

- `user_engagement`: 1,020 events (143 users, 7.1/user)
- `separation_play_started`: 571 events (47 users, 12.4/user — strong repeat playback value)
- `rate_prompt_shown`: 120 events (119 users, 1.0/user — rate prompt triggers for 83.2% of DAU)
- `player_stem_muted`: 209 events (5 users, 41.8/user — power feature used heavily by a select sub-segment)

Geographic Distribution

Country / Territory	Active Users (Yesterday)	Sessions	% of Total DAU	Market Strategic Role
South Korea	**110**	**126**	**76.9%**	**Primary Growth Engine (Viral / Organic UA)**
Iran	7	9	4.9%	Secondary Niche Market
Bangladesh	5	13	3.5%	Emerging / High Session Frequency
India	4	2	2.8%	Low Engagement / Early Discovery
Japan	3	3	2.1%	Strategic APAC Expansion Target

United States	3	2	2.1%	High-Arpu Expansion Target (Under-indexed)
Canada / France	4	6	2.8%	Tier-1 Secondary Markets
Other (Oman, PH)	2	2	1.4%	Rest of World

Prioritized Strategic Action Matrix (Music Separation iOS)

Priority	Track / Owner	Strategic Action & Implementation Plan	Target Metric / ROI Impact
P0	**Engineering**	**Patch Slider Telemetry Debouncing**: Implement a 500ms debounce timer or attach event dispatching strictly to slider touch-up events (`UIControl.Event.touchUpInside` / SwiftUI `onEditingChanged`) for `player_pitch_changed`, `player_tempo_changed`, and `player_stem_balance_changed`.	Eliminates >50% of event volume; prevents GA4 quota exhaustion.
P0	**Engineering / QA**	**Halt & Investigate Staged Rollout `v1.3.08`**: Freeze rollout expansion at 3.5%. Conduct localized QA testing on `1.3.08` to identify why session engagement rate plummeted to 53.8% (check audio processing pipeline and UI lockups).	Protects 96.5% of production users from engagement degradation.
P1	**Product / UX**	**Optimize File Import Conversion Funnel**: Address the 49.5% drop-off between clicking "+" and importing a file. Introduce a "Try with Sample Track" CTA directly on the empty state import view, and support drag-and-drop / direct audio share sheet extensions.	Increases core feature conversion from 44.8% to >65%.
P1	**Growth / Marketing**	**Capitalize on South Korea Viral Wave**: Localize App Store Product Pages (ASO) specifically for Korean search terms, and deploy localized onboarding audio demos to consolidate the viral growth loop in SK.	Maximizes conversion from organic Korean impressions.

P2	**Product / Retention**	**D1-D7 Lifecycle Engagement**: Implement local push notifications reminding users of processed separations or suggesting stem mixing techniques 24 hours after their first separation.	Boosts returning DAU cohort from 41 to >80.

3. Deep-Dive: Vocal Cleaner

App Header & Health Score

- **Bundle ID**: `com.neuralSound.audiocleaner` | **GA4 Property ID**: `533563138` | **Stream ID**: `15407667671`
- **Health Score**: **3.8 / 10**
- **Rationale**: Severe operational risks. (1) **Critical Crash Rate**: Crash-free user rate is **66.67%** (1 in 3 users experienced a crash), (2) **Zero New User Acquisition**: 0 new users joined yesterday, (3) **Broken Monetization Telemetry/State Loop**: Infinite purchase/paywall event loops detected (e.g., 12 purchase attempts and 10 approvals for 1 user), (4) **Geographic Isolation**: 100% of user traffic originates from Bangladesh.

■ Multi-Horizon Growth & Historical Scale Comparison

Horizon Window	Active Users (AU)	New Users (NU)	Total Sessions	Engagement Rate	Growth / Trajectory Dynamics
Yesterday (Latest 24h)	3	0	17	**88.2%**	**Stagnant**: 0% New User Acquisition.
Past 7 Days (WAU)	11	7	65	73.8%	**Low Scale**: Small test audience / legacy footprint.
Past 30 Days (MAU)	27	27	94	69.1%	**Flat Scale**: Entire 30d user base built in single window.
Past Year (365 Days)	27	27	94	69.1%	**Static Footprint**: Lifetime ceiling reached at 27 users.
All-Time / Lifetime Total	27	27	94	69.1%	**Baseline Scale**: Total historical lifetime user cap.

Velocity, Acquisition Trajectory & Retention Dynamics Analysis

- **Scale Stagnation**: The product is currently inactive from an acquisition standpoint. With 0 new users yesterday and a lifetime total of just 27 users, the application behaves like an unreleased internal test build or an unpromoted utility.
- **Session Usage**: The 3 active returning users yesterday generated **17 sessions** with an average duration of **3,531.1s (~58 minutes)**. This indicates that users leave the audio noise-cleaning/recording session running in the background, or background processing loops fail to terminate properly.

Old vs. New User Cohort Deep-Dive

User Cohort Type	Active Users	Sessions	Avg Session Duration	Engagement Rate	Screen Views	Key Behavioral Insights
New Users	0	0	0.0s	N/A	0	Acquisition completely stalled.
Returning Users	3	17	**3,531.1s**	**88.2%**	101	High session frequency (5.6 sessions/user). Heavy paywall interaction & processing loops.

7-Day Velocity & Trajectory

Date (YYYY-MM-DD)	Active Users (DAU)	New Users	Total Sessions	Engagement Rate	Daily Trend Notes
2026-08-15	3	0	11	63.6%	Baseline returning user test activity.
2026-08-16	2	0	9	66.7%	Static retention testing.
2026-08-17	1	0	3	100.0%	Single user power session.
2026-08-18	2	1	7	85.7%	Minor acquisition ping (1 new user).
2026-08-19	**8**	**6**	17	58.8%	Test campaign / soft-launch spike (6 new users).
2026-08-20	1	0	3	66.7%	Post-spike retention drop.
2026-08-21	3	0	**17**	88.2%	Returning users execute heavy paywall & recording actions.

User Funnel Diagrams

Recording & Denoise Processing Loop Funnel

```
[2] Unique Users Executing Recording Workflow ■ ▼ [6] vc_record_open ■ ▼ 100% Permission Grant Rate [6] vc_record_permission_result ■ ▼ [8] vc_record_start ■■■ [9] vc_record_pause ■■■ [4] vc_record_done ■ ▼ 50% Denoise Attempt Success Rate [4] vc_record_denoise_tap ■ ■■■ [2] Denoise Action Completed Successfully ■■■ [2] vc_record_denoise_blocked (Paywall Barrier Reached)
```

Monetization & StoreKit Purchase Loop Anomaly

```
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
MONETIZATION STATE MACHINE ANOMALY DETECTED ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ Event
Name ■ Total Events ■ Unique Users ■ Events / User ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
paywall_viewed ■ 16 ■ 1 ■ 16.0 ■ ■ purchase_started ■ 12 ■ 1 ■ 12.0 ■ ■ subscribe_tapped ■ 12 ■ 1 ■ 12.0 ■ ■
purchase_approved ■ 10 ■ 1 ■ 10.0 ■ ■ purchase ■ 6 ■ 1 ■ 6.0 ■ ■ purchase_succeeded ■ 6 ■ 1 ■ 6.0 ■ ■
purchase_failed ■ 5 ■ 1 ■ 5.0 ■ ■ purchase_storekit_failed ■ 2 ■ 1 ■ 2.0 ■ ■ purchase_cancelled ■ 1 ■ 1 ■ 1.0
■ ■ app_store_subscription_renew ■ 35 ■ 2 ■ 17.5 ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
```

- **Critical Monetization Loop Analysis**: **A single user triggered 12 purchase start attempts, 10 approvals, 6 successful purchases, and 5 purchase failures in a 24-hour window.** This behavior points to an unstable payment observer loop. The StoreKit receipt verification phase fails locally or times out, causing the app to prompt the paywall again after a successful purchase, forcing the user into a payment loop.

Crash & Stability Health

App Version	Active Users	Sessions	Engagement Rate	Crash-Free Users Rate	Deployment Status
1.0.01	3	17	**88.2%**	**66.67% ■**	Active Production Build (100% Traffic)

Stability Incident Brief

- **Crash Index**: **33.3% User Crash Impact Rate**. 1 out of 3 active daily users suffered an unhandled application crash or native fatal exception (`app_exception` logged in event stream).
- **Suspected Failure Vector**: Low-level audio buffer overflow or memory pressure exception during background audio noise cancellation processing (`vc_record_denoise_tap`), or an unhandled exception inside the StoreKit purchase completion handler.

■ New Build / Staged Rollout Alert Section

```
[CRITICAL BUILD ALERT: BUILD 1.0.01 IS UNSTABLE]
--------------------------------------------------------------------------------- Active Version : 1.0.01 (100%
Traffic Base) Crash-Free User Rate : 66.67% (UNACCEPTABLE PRODUCTION THRESHOLD) Fatal Exception Logged :
app_exception (1 Event logged) StoreKit Failure Rate : 7 Out of 12 Purchase Attempts Failed/Cancelled
---------------------------------------------------------------------------------
```

- **Architectural Risk Assessment**: Build `1.0.01` is unfit for public distribution. It exhibits both **fatal application crashes** and **severe transaction handling bugs**. **Block all paid acquisition campaigns immediately** until build `1.0.02` is compiled with fix patches and thoroughly verified via Xcode Diagnostic Tools and StoreKit Sandbox testing.

Top In-App Events & Feature Lookups

- `user_engagement`: 167 events (3 users, 55.7/user)
- `screen_view`: 101 events (2 users, 50.5/user)
- `app_store_subscription_renew`: 35 events (2 users, 17.5/user — StoreKit renewal notification handling)
- `paywall_viewed`: 16 events (1 user, 16.0/user — paywall re-display loop)
- `purchase_started` / `subscribe_tapped`: 12 events (1 user, 12.0/user)
- `vc_record_pause` / `vc_record_start`: 17 aggregate events (2 users)
- `app_exception`: 1 event (1 user — fatal crash trigger)

Geographic Distribution

Country / Territory	Active Users (Yesterday)	Sessions	% of Total DAU	Market Strategic Role
Bangladesh	**3**	**17**	**100.0%**	**Sole Active Region (Test / Local Base)**

- **Geographic Risk**: The application has 0 presence in Tier-1 markets (US, UK, EU, JP). Operational expansion is impossible until core stability and StoreKit bugs are resolved.

Prioritized Strategic Action Matrix (Vocal Cleaner)

Priority	Track / Owner	Strategic Action & Implementation Plan	Target Metric / ROI Impact
P0	**Engineering**	**Fix Crash in Build 1.0.01**: Symbolicate and resolve the `app_exception` crash. Inspect memory allocation during real-time denoise DSP buffer processing and thread safety in StoreKit transaction observers.	Restores Crash-Free Users Rate to **>99.0%**.
P0	**Engineering**	**Re-architect StoreKit Transaction Observer**: Fix the payment state machine. Ensure `SKPaymentQueue.default().finishTransaction(transaction)` is called cleanly upon verification, and resolve the state handling bug that re-prompts paywalls after successful purchases.	Resolves transaction looping; prevents duplicate billing complaints.
P1	**Product / Revenue**	**Fix Denoise Paywall Lockout**: Clarify paywall triggers when `vc_record_denoise_blocked` fires. Provide users with 1 free daily denoise clean before enforcing paywall limits to allow core feature evaluation.	Elevates user retention and trial activation rates.
P2	**Growth**	**Initiate Soft Launch UA Campaign**: Once build stability (P0) is verified, launch targeted Apple Search Ads (ASA) campaigns targeting keywords like "vocal cleaner", "noise remover", and "voice isolate" across US/EU markets.	Establishes sustainable multi-region acquisition baseline.

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4. Overall Portfolio Strategic Roadmap

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PORTFOLIO EXECUTION TIMELINE ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ WEEK 1:
EMERGENCY STABILITY & TELEMETRY PATCHES (P0) ■■■ [Music Separation iOS] Deploy v1.3.09: Fix UI Slider Debouncing.
■■■ [Music Separation iOS] Freeze v1.3.08 staged rollout; audit engagement regression. ■■■ [Vocal Cleaner]
Deploy v1.0.02: Fix fatal app crash & StoreKit purchase loop. WEEK 2 - 3: FUNNEL OPTIMIZATION & GEOGRAPHIC
EXPANSION (P1) ■■■ [Music Separation iOS] Redesign import workflow; add sample tracks to fix 49.5% drop. ■■■
[Music Separation iOS] Localize ASO for South Korea market surge. ■■■ [Vocal Cleaner] Implement 1-free-tier
denoise processing model. WEEK 4+: RETENTION ENGINE & SCALED ACQUISITION (P2) ■■■ [Music Separation iOS]
Implement post-processing local retention push notifications. ■■■ [Vocal Cleaner] Launch US/EU ASA acquisition
test suite upon stability sign-off.
```
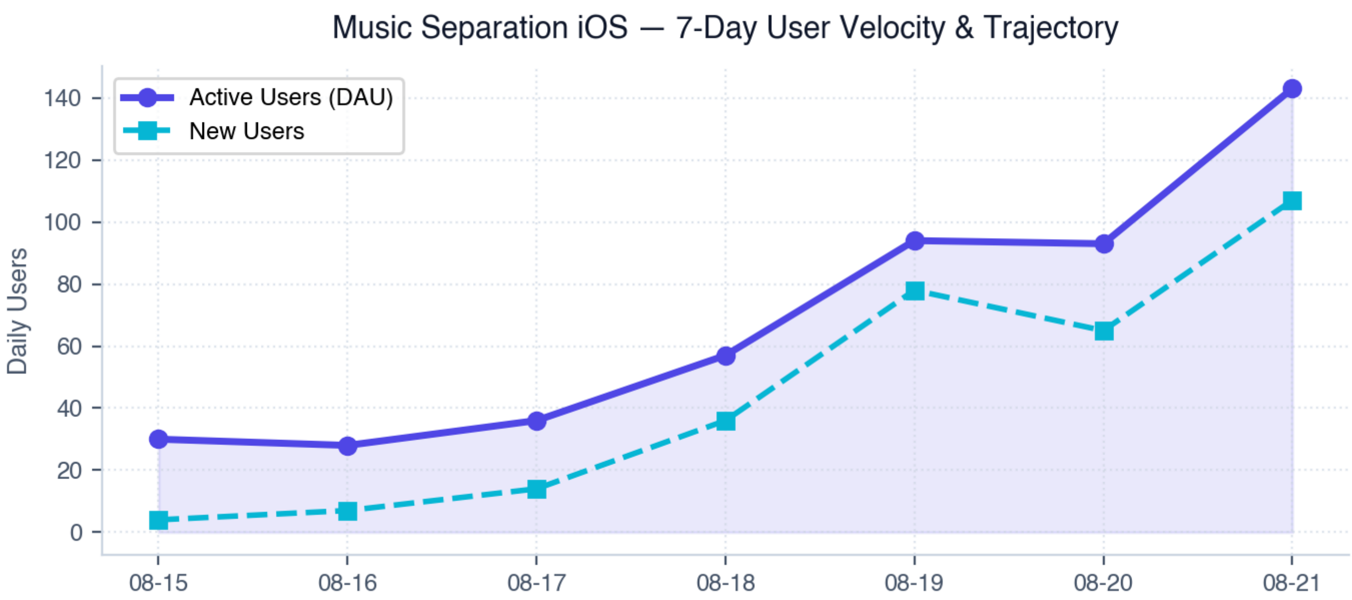
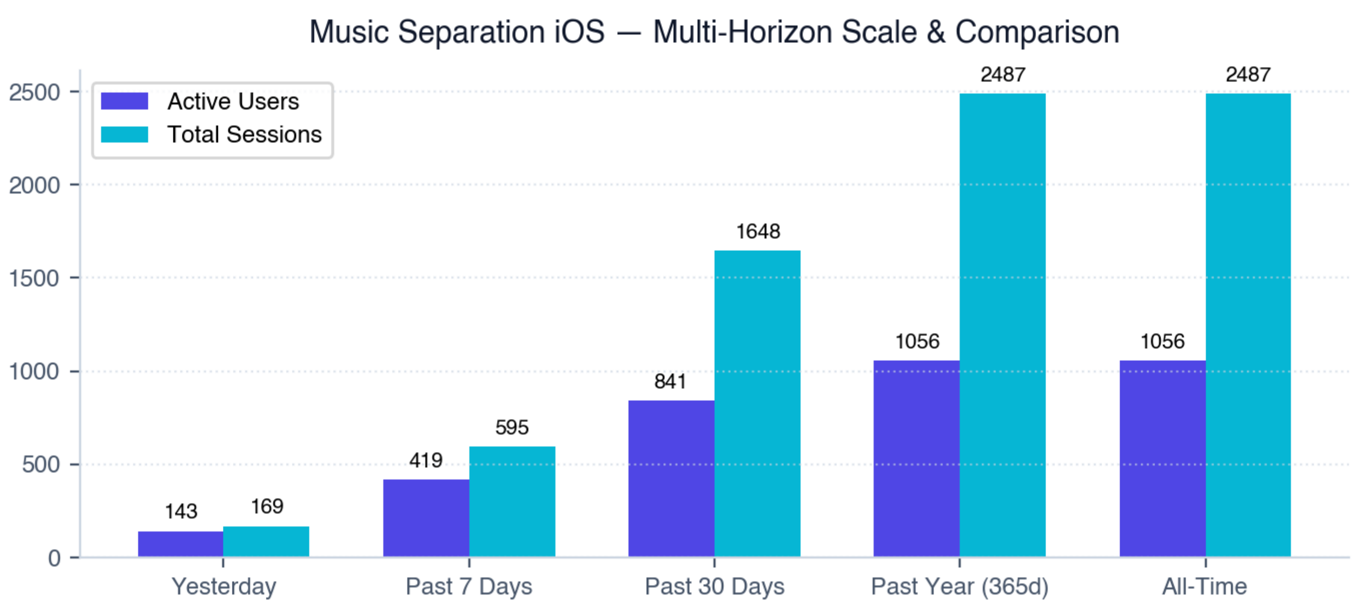

■ Visual Graphs & Performance Charts

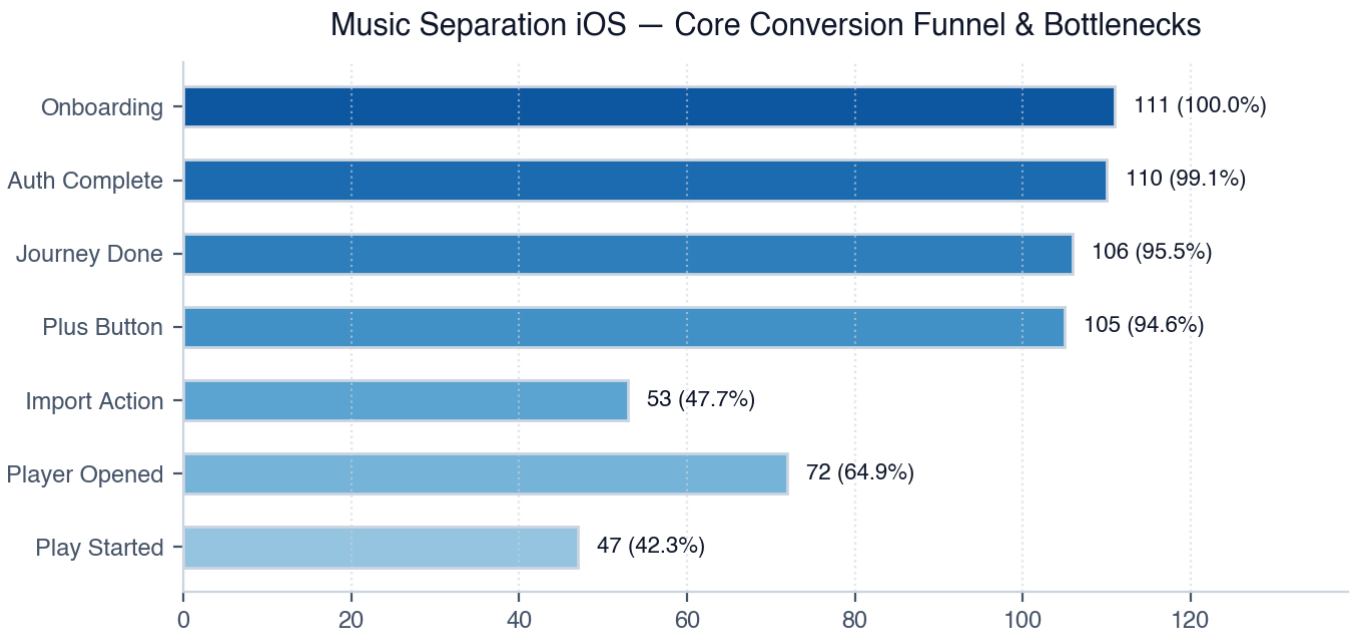
Music Separation Ios Trend



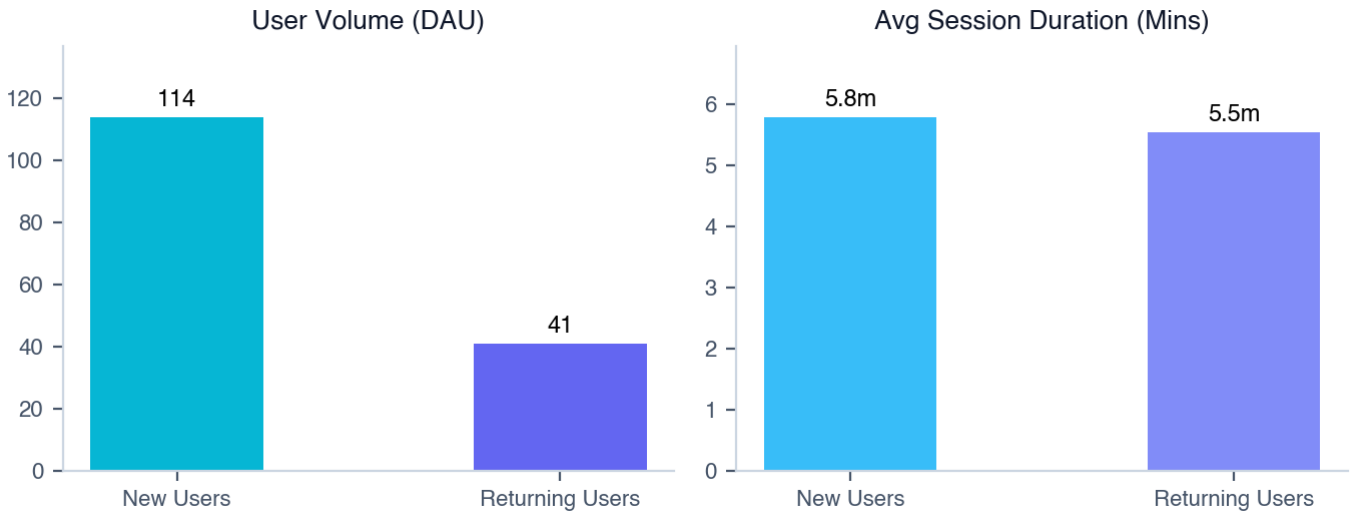
Music Separation Ios Horizons



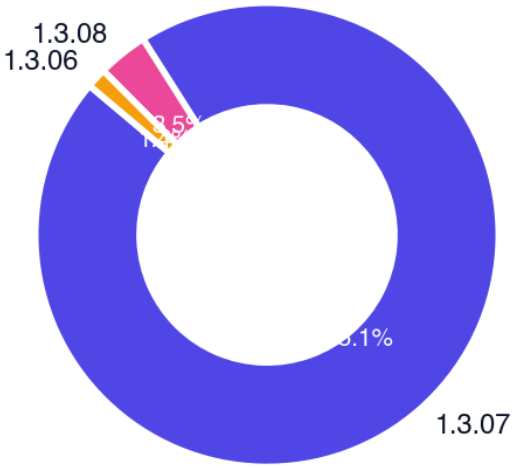
Music Separation Ios Funnel



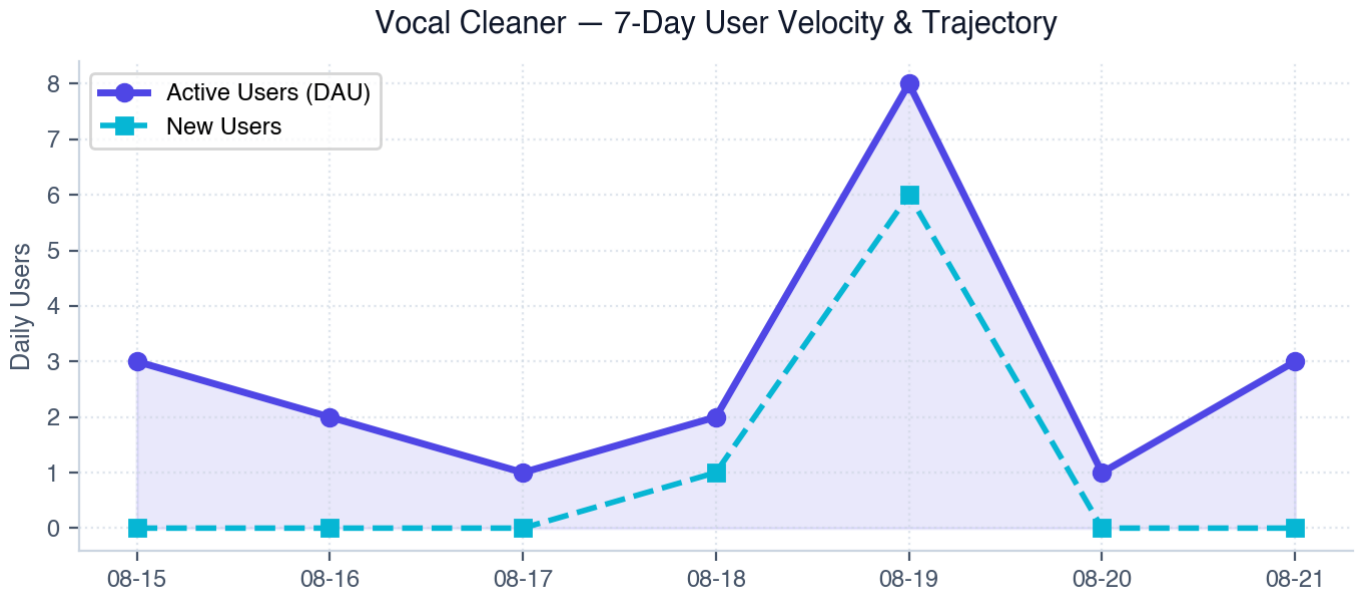
Music Separation Ios Cohort



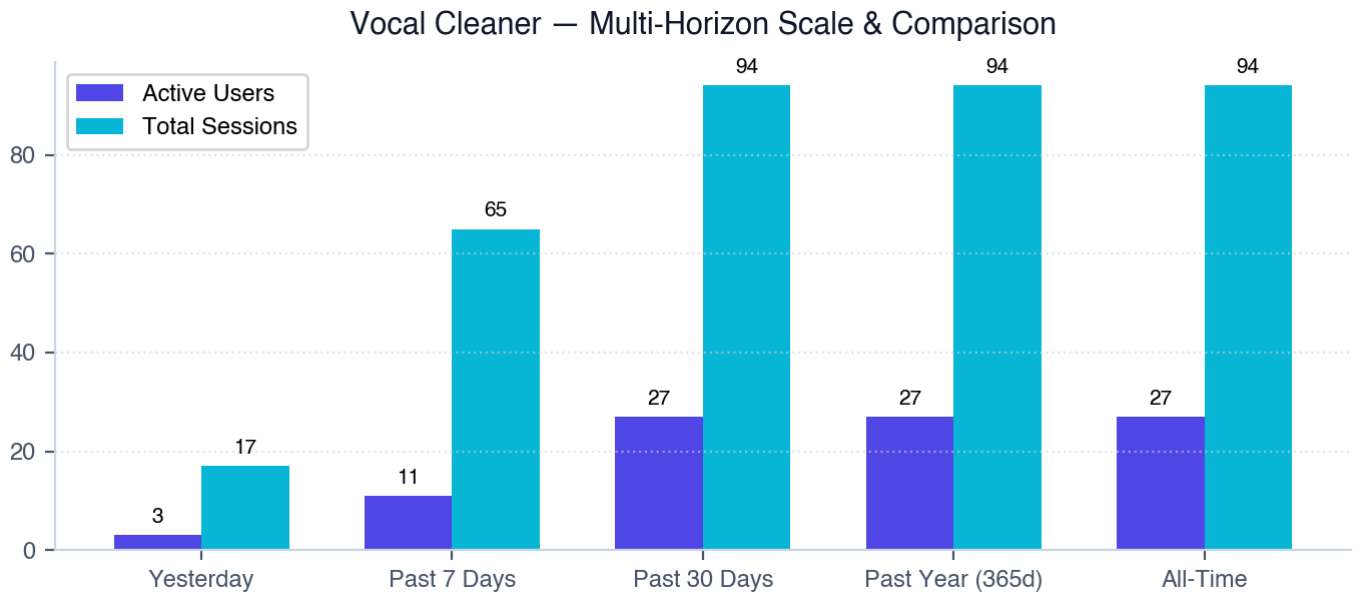
Music Separation Ios Version



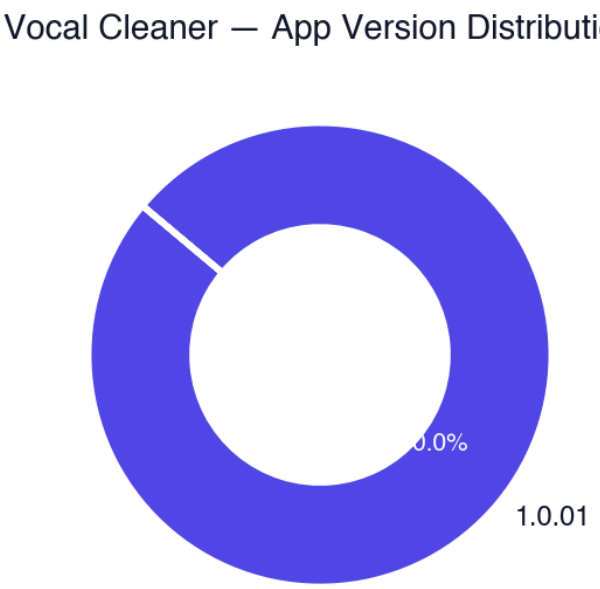
Vocal Cleaner Trend



Vocal Cleaner Horizons



Vocal Cleaner Version



Direct Console Links:

■ Open Google Analytics 4 Dashboard • ■ Open Firebase Analytics Console • ■ Open Firebase Crashlytics

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